

Video Game Programming: Story and Levels

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Game story

- Narrative plays a fundamental role in games
- Some perceive gaming and storytelling as incompatible
- Several games such as Zelda, BioShock, and Blade Runner suggest the opposite
- For sure, a game story (nothing happens if player does not make a choice) is different from a movie story (sit and watch action)



Screenshot from Blade Runner (game)

Story and genre

- Story has a more prominent role in some genres than others
 - In adventure games, story is essential
 - In role-playing and action-adventure games, story is important
 - In other genres, e.g., racing or fighting games, story is less relevant
- In the past there were games with almost no story
- Nowadays all popular genres use some fictional elements
- E.g., Star Wars Episode I: Racer has only a connection with the movie story, but without it the game would be just a racing game

Definitions

- Narrative video games: those in which story plays a significant role
- Narrative (or plot): succession of events
- Story: chronological order of the events themselves (story can also more generally refer to fictional worlds)
- Text: verbal or visual representation of events
- Narration: act of telling or writing (single telling is rare in games)
- Fiction: events that have not occurred in real life (almost all games stories are fictional)

Story elements

- A story usually relies on different elements:
 - Settings (environments, objects,...)
 - Characters (protagonists, antagonists,...)
 - Events and their structure
 - Theme (central concern that comes up again and again)
 - Genre (how to play, tone and mood)
- Settings and characters can appear on their own (without a narrative), so that players can imagine how they fit into the fictional world or relate them to an external story
- Storytelling can be addressed from three perspectives: fictional world, mechanics of the narrative, and reception (the way in which players experience the story)

Fictional world

- It includes settings and characters
- World's objects and space are organized around the act of play
- In general, only objects related to gameplay will be usable
- Games indicate which objects are relevant in different ways, e.g., changing cursor, making them more alive or attractive than others
- Some games (e.g., GTA V) try to make all objects alive and the world creates the illusion to be active even without player

Gamespace

- Gamespace is the game world part that is the setting for the gameplay
- Gamespaces are not realistic but reductive: they reproduce some features of real world but with their own rules to facilitate gameplay
- Gamespaces are also meant to reduce the processing power required by a computer to run the game
- Single-player games often have landscapes that seem open but are restrictive to orchestrate players' movement in one direction
- Multi-player landscapes should be open so that all players can move freely, and no player starts with an advantage

Cutscenes

- Cutscenes use in games is disputed, but they have several roles:
 - They introduce a central narrative tension, e.g., in many arcade games, animated cut-scenes introduced the central conflict of the game
 - They shape the narrative in a certain direction, e.g., to ensure that the game protagonist makes certain choices
 - They compensate for missing game narrative, e.g., between two game scenes
 - They associate the game with contemporary cinema aesthetics, e.g., to build a richer game experience through an emotional event
 - They provide players with information, e.g., the camera may swoop through a scene showing the placement of enemies and objects

Characters

- In other media, characters are the people that the media is about
- Games introduce the possibility to play some characters, and that other characters will react to players
- Therefore, characters in games are the people who are making action happen and thus producing different stories
- Main characters change and evolve over the course of a game
- There can be a relationship between some characters' mechanical abilities in the game and their position in the story

Character functions

- Characters can be organized in types according to the extent to which players can interact with them:
 - **Stage** characters are part of the scenario, moving around but with no personality or function; players cannot interact with them;
 - **Functional** characters are like stage characters but with a general function in the game;
 - **Cast** characters are characters with a particular function in the game related to the story; they can have personality and their own agendas;
 - **Player** characters are the characters controlled by the player (except in cutscenes)

Player characters

- Player characters can in turn be organized in three types:
 - Avatars are a non-intrusive representation of players
 - Actors are part of a story (or have a story, albeit minimal sometimes)
 - Roleplaying characters have very different abilities that we can raise according to players' performance
- Avatars usually cannot be seen, and the game is in first person, so that players can merge with them (e.g., in Unreal)
- Actors are usually seen in third person (e.g., in Tomb Raider)
- Roleplaying characters are usually customized by players and are seen in first or third person (e.g., in MMORPGs)

Character archetypes

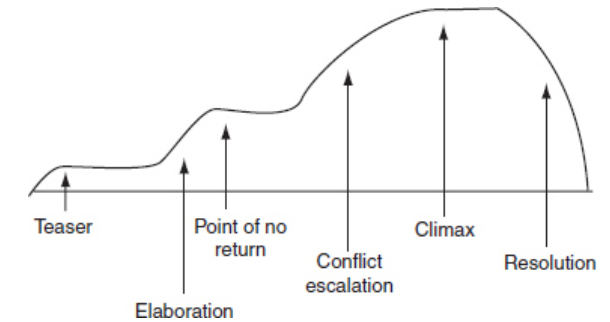
- Some characters have peculiar features (e.g., the hero, the helper, the best friend of players' character, the person to save)
- Such features are typically common in different stories (in games as well as in other media)
- Archetypes help creating stories by relying on familiar dynamics
- However, archetypes can also be used to subvert expectations (e.g., friends becoming enemies or vice versa)
- Some archetypes have predefined looks that communicates information about the character and the world that they are in

Mechanics and players' agency

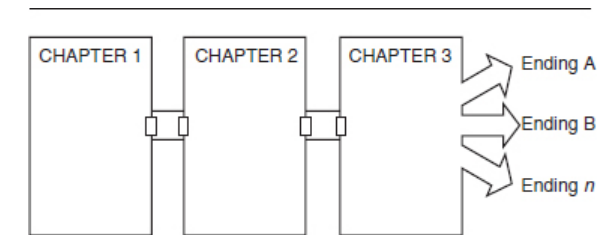
- Mechanics are about organizing narrative action
- One of the most difficult tasks is combining a playing experience that feels free with the constraints of a narrative structure
- Players' agency is their ability to make choices
- Game programmers should determine how the choices players make affect the story and the game
- Problem of linearity: how to get a player to move through a game in one specific way

Traditional / interactive fiction

- Traditional fiction goes only one way toward a resolution
- Applying this to games, players will not be able to do anything, as any deviation would ruin the planned emotional curve
- In interactive fiction, players solve puzzles within each chapter
- Successive chapters work in a cumulative way: at the end of each, a climax or resolution is provided, usually after a boss fight
- The key is to make players feel that they are contributing to create a plot



Classic linear fiction



Interactive fiction

Branching

- One way to do this is to organize narrative action with branching
- Branching is the existence of multiple paths in a narration
- When players make some choices, they might see different outcomes
- Branching leads to the problem of managing the exponential growth of nodes in a narrative
- Sometimes branches can come back together (plot bottlenecks), but the players should not have the impression that their choices are meaningless

Environmental storytelling

- Environmental storytelling is when information about the story is in the environment that players encounter, instead of having a character say it out loud
- Murder mysteries are a good example of environmental storytelling: readers must understand the story of what happened from the scene of the crime
- Players are free to explore the scene as they prefer, but the story behind the mystery is the same (even if players' actions can lead to different endings)

Quests

- Game programmers can also manage smaller sequences of events that make up the arc of a game's plot
- Quests are small missions that players can or must perform
- Good quests create opportunities for storytelling
- Bad quests fail to integrate the storytelling elements in a particular game
- There are some classical quests (e.g., delivering a letter / box to one NPC) that can be promising to develop the story or introduce new characters

Narrative structures

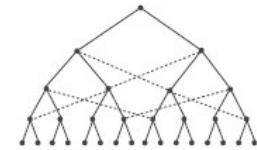
- Tree: paths are unidirectional, and progression produces a plot
- Vector with side branches: there is one main story that progresses chronologically, but players are encouraged to make small detours
- Hidden story structure: it has a fixed story that the players slowly uncover as they progress
- Braided plot: it allows the player to experience different perspectives on an unfolding narrative by changing the protagonist and location



1). Complete graph structure



2). Network structure



3). Tree structure



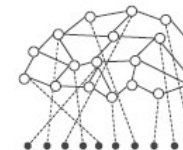
4). Vector with side branches



5). Maze structure



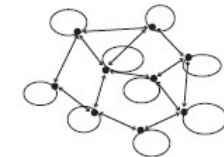
6). Directed network



7). Hidden story structure



8). Braided plot



9). Action space, epic wandering and story-world

Level flow

- Games that are too simple or too easy are boring
- Games that are too complicated or too difficult cause anxiety
- Games that are well balanced encourage a state called flow in players, where they're engaged and focused on play
- Game balance is the practice of finding the right balance of difficulty and reward to keep players engaged
- Level design has a huge impact on flow, as it helps designers determine how to keep the player engaged without either boring or overwhelming them

Level design

- Key questions: What's the overall concept of the level? Where does it take place? What happens over the course of the level?
- It's important to research real-world counterparts for the levels the players are making
- Designers can create floor plans drawing out the physical space
- It's important to mark out where specific gameplay interactions will happen in the level
- Elements include ambushes, explosions, use of light... anything that will trigger when a player interacts with it
- Objects can be used to fill space or tell stories

Tools for stories and levels

- Storyboards: a tool used to show different shots or scenes in a story, it is an important tool for communicating an idea visually
- Game Design Document: it is a record of every concept, asset, and mechanic that is going to go into a game; it is the most important tool for a team to collaborate on, because it gives everyone information about the state of the game
- Terrain tools and environmental objects: some game engines offer tools to create level geometry, such as terrains

Open research

- Ludology (study of game structure or gameplay) versus narratology (the study of game narrative)
- Interaction versus narrative (interactive storytelling, natural language use as in Façade in fig.)
- Transmedia games (games whose story develops in other media)



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